

Resilience Practice

ENGCV370

Dr Megan Boston



Learning Objectives

- Practice applying Resilience
 - Robustness
 - Redundancy
 - Resourcefulness
 - Rapidity

Topics Covered

- Resilience
 - Definition of Resilience
 - Resilience Triangle
 - Dimensions of resilience

Resilience

- After the 2016 Kaikoura Earthquake, severe ground slips and landslides destroyed major sections of State Highway 1 between Christchurch and Picton. Initially, communities along the coast were isolated while repairs were made to State Highway 1 and other roads. An abbreviated timeline of the repairs is given.
 - 14 November 2016: Kaikoura Earthquake, State Highway 1 between Picton and Christchurch is closed. The inland road between Kaikoura and Christchurch is closed.
 - 19 December 2016: The inland road reopens between Kaikoura and Christchurch
 - 21 December 2016: State Highway 1 between Kaikoura and Christchurch is reopened
 - 15 December 2017: State Highway 1 reopens from Kaikoura to Picton
- Reopening the different sections of road around Kaikoura increased the ability for people to travel easily between Christchurch and Picton and out of Kaikoura.



Resilience curves

- The Waiho Bridge in Franz Josef is an important link on SH6 providing access across the Waiho River connecting communities along the west coast of the South Island. The bridge was washed away and destroyed on 26 March 2019 during a massive rainstorm. Without the bridge, travel time between Franz Josef and Fox Glacier increased from 25 minutes to 14 hours with vehicles required to travel around Aoraki/Mt Cook. If we assume the following:
 - Prior to the bridge collapse the number of vehicles that travel between Franz Josef and Fox Glacier across the Waiho bridge on SH6 was 1250 each day,
 - During the bridge repair (26 March to 12 April 2019) the number of vehicles that travel across is 0 per day,
 - After repairs are complete the number of vehicles crossing the bridge returns to the original numbers of vehicles per day,
- What does the resilience curve look like that best describes the number of cars able to travel between Franz Josef and Fox Glacier on SH 6?





Milford Rd State Highway 94. Photo: NZTA

SOUTHLAND FLOODS 2020

1100 mm of rain over 60 hr period

~200 tourists in Milford Sound

~200 staff in Milford Sound

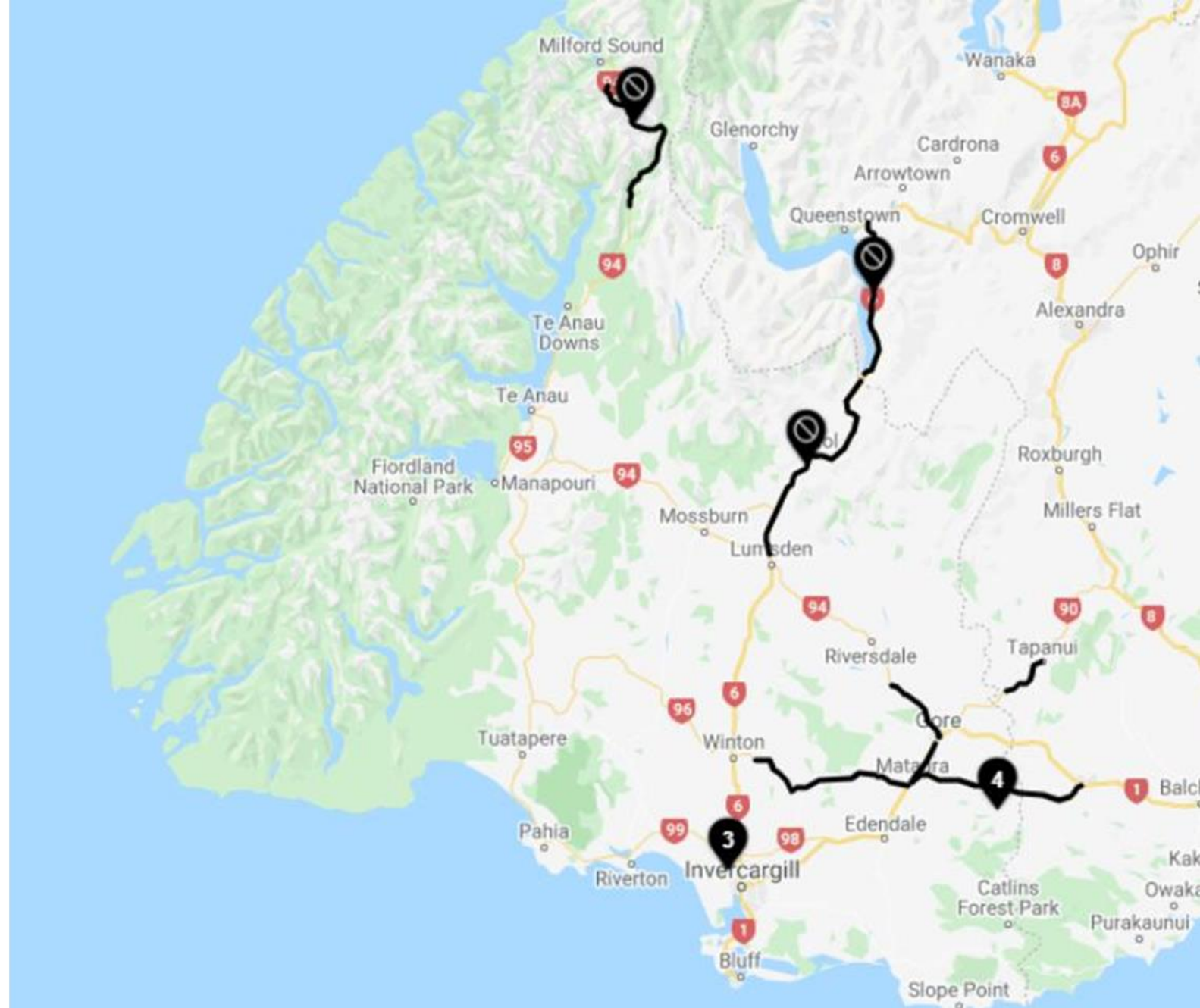
~70 people stranded along Milford road

100+ people trapped in Great Walk and backcountry Huts

6000 people evacuated from Gore, Wyndham, and Mataura

Milford Road closed for weeks, partial reopening to convoys only

SOUTHLAND FLOODS 2020



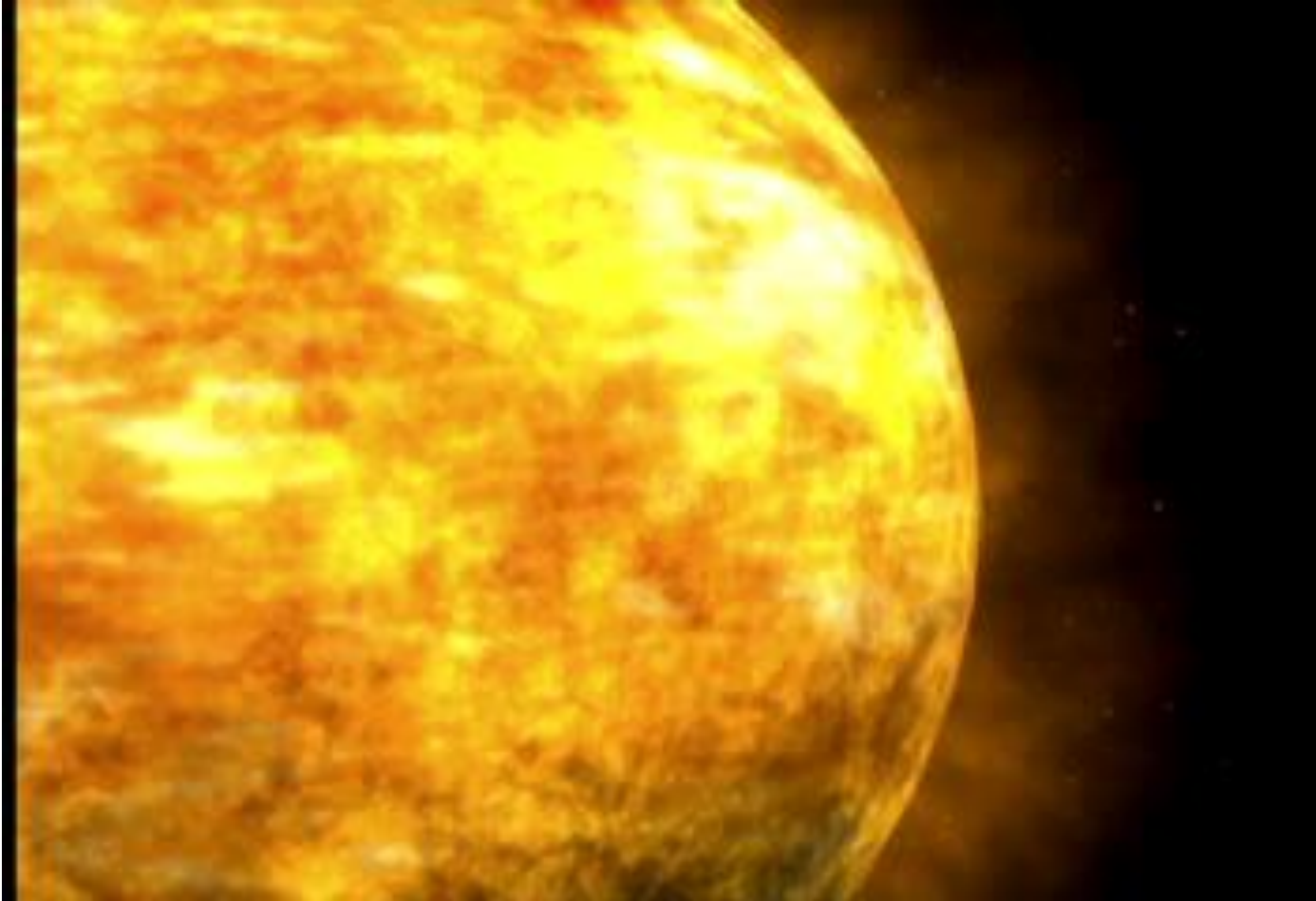
Road Closures due to Southland Flooding, Photo:NZTA

Carrington Solar Flare

August 28-September 2, 1859

- 
- Those who happened to be out late on Thursday night had an opportunity of witnessing another magnificent display of the auroral lights. The phenomenon was very similar to the display on Sunday night, though at times the light was, if possible, more brilliant, and the prismatic hues more varied and gorgeous. The light appeared to cover the whole firmament, apparently like a luminous cloud, through which the stars of the larger magnitude indistinctly shone. The light was greater than that of the moon at its full, but had an indescribable softness and delicacy that seemed to envelop everything upon which it rested. Between 12 and 1 o'clock, when the display was at its full brilliancy, the quiet streets of the city resting under this strange light, presented a beautiful as well as singular appearance.
 - [Baltimore American and Commercial Advertiser](#)
 - 3 September 1859

Carrington Solar Flare

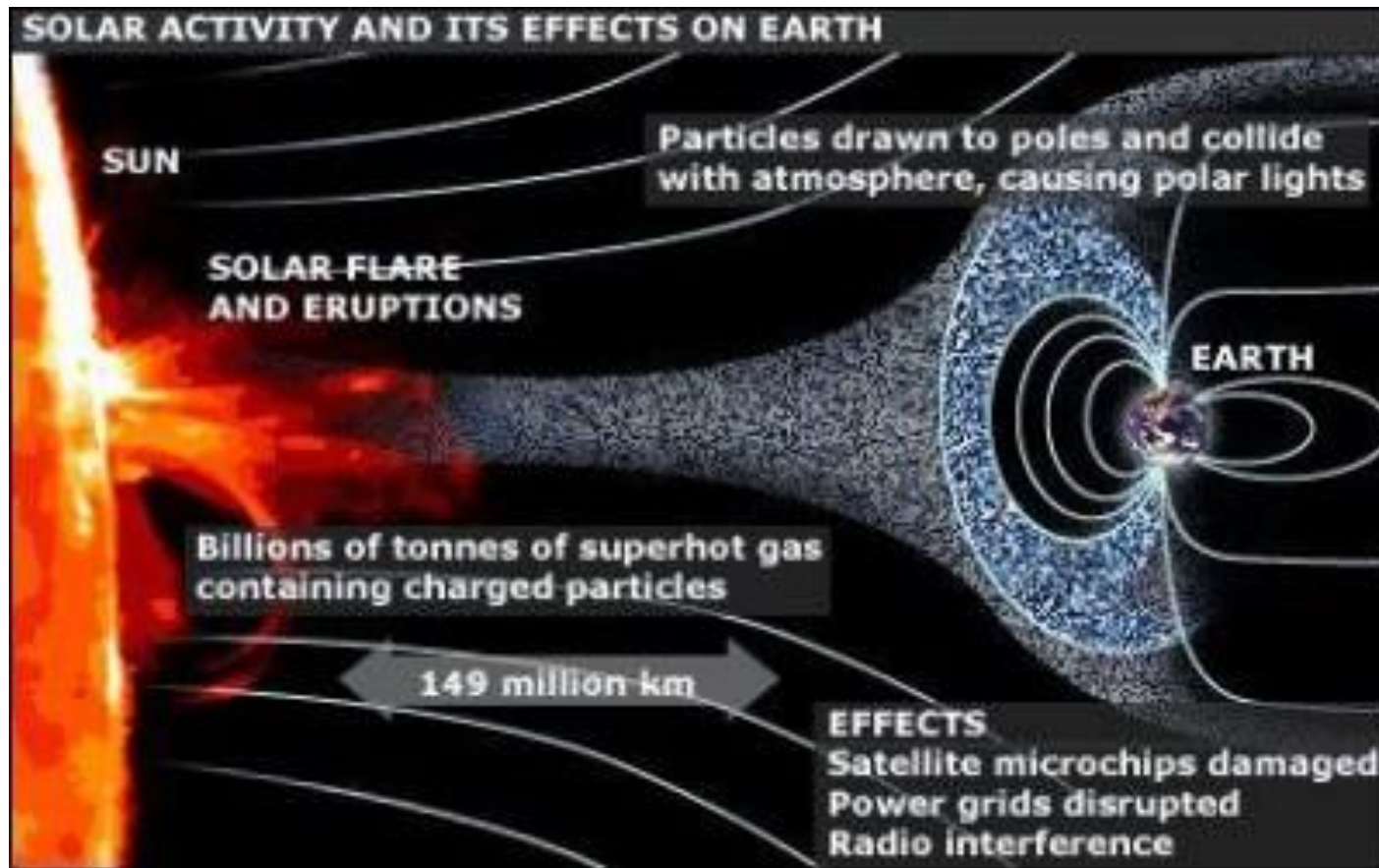


Coronal Mass Ejection

NASA/ Walt Feimer

- Geometric solar storm
- Knocked out telegraphs

If it happened today...



- “We are estimating that 20 to 40 million people might be without power from anywhere up to one, even two years. That has to do with the critical issue of replacement transformers. That number of people without power could result in an economic cost somewhere between \$.6 trillion to \$2.6 trillion.”

-Neil Smith, Exposure Management

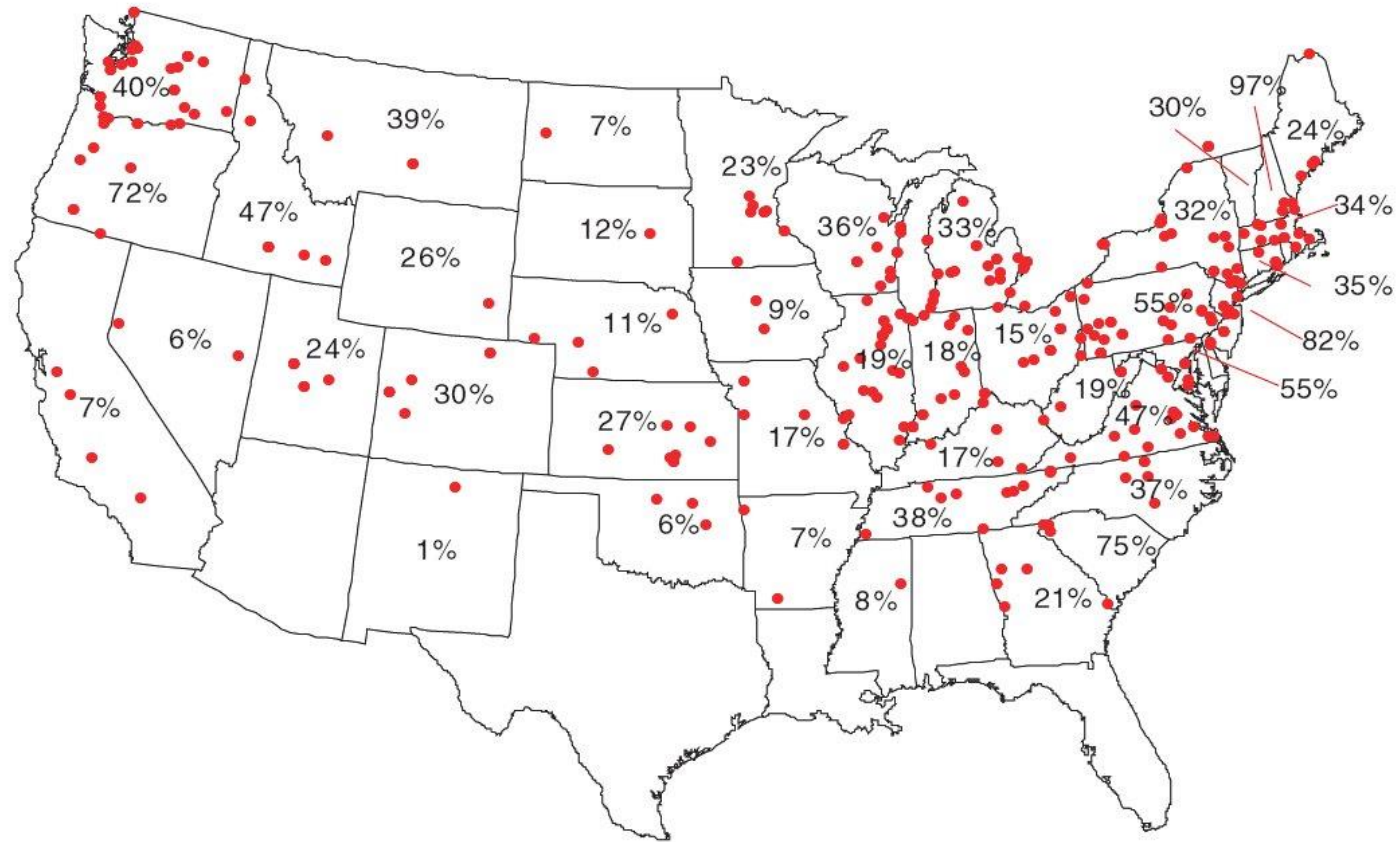
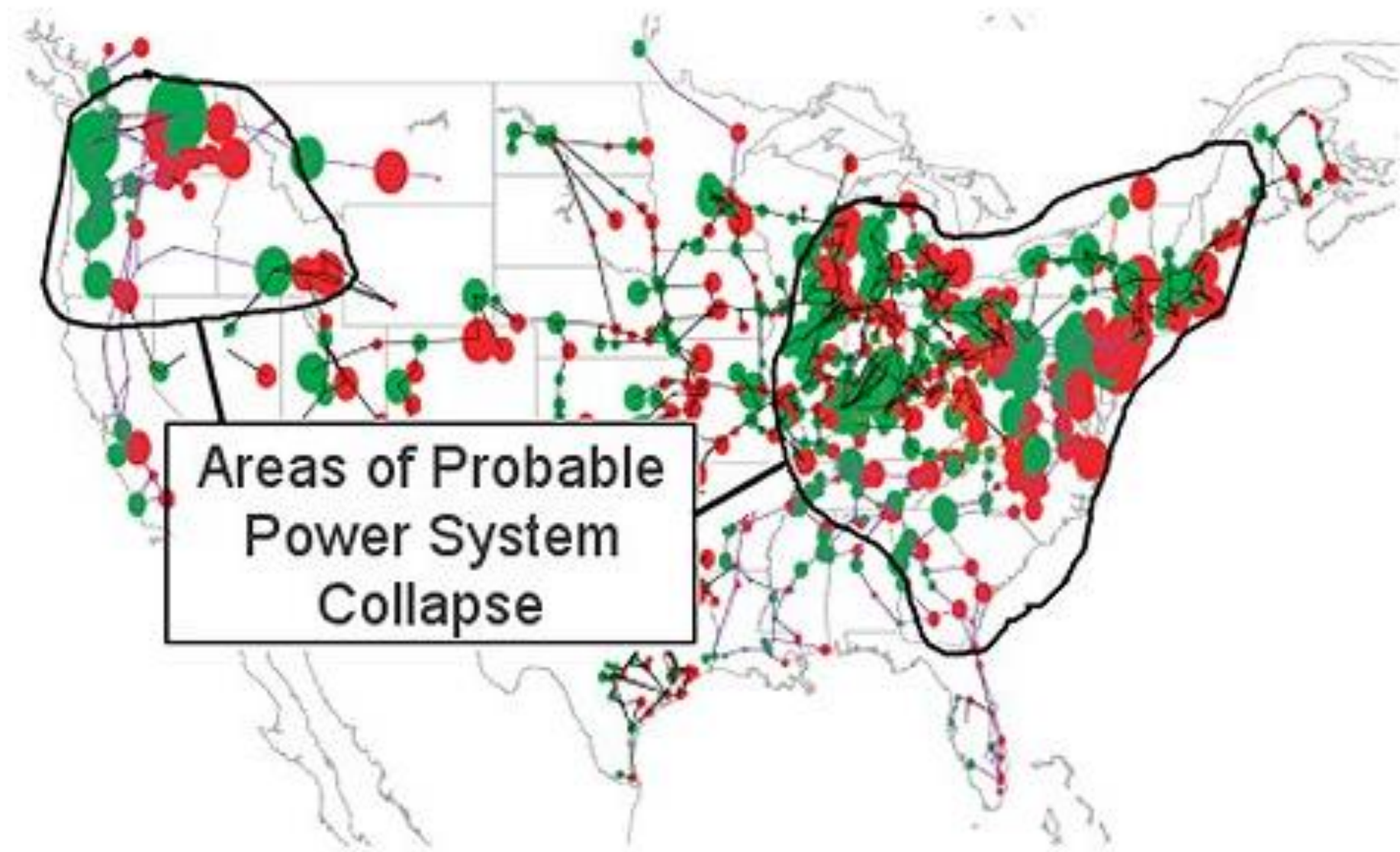


FIGURE 7.2 A map showing the at-risk EHV transformer capacity (estimated at ~365 large transformers) by state for a 4800 nT/min geomagnetic field disturbance at 50° geomagnetic latitude. Regions with high percentages of at-risk capacity could experience long-duration outages that could extend multiple years. SOURCE: J. Kappenman, Metatech Corp., “The Future: Solutions or Vulnerabilities?,” presentation to the space weather workshop, May 23, 2008.



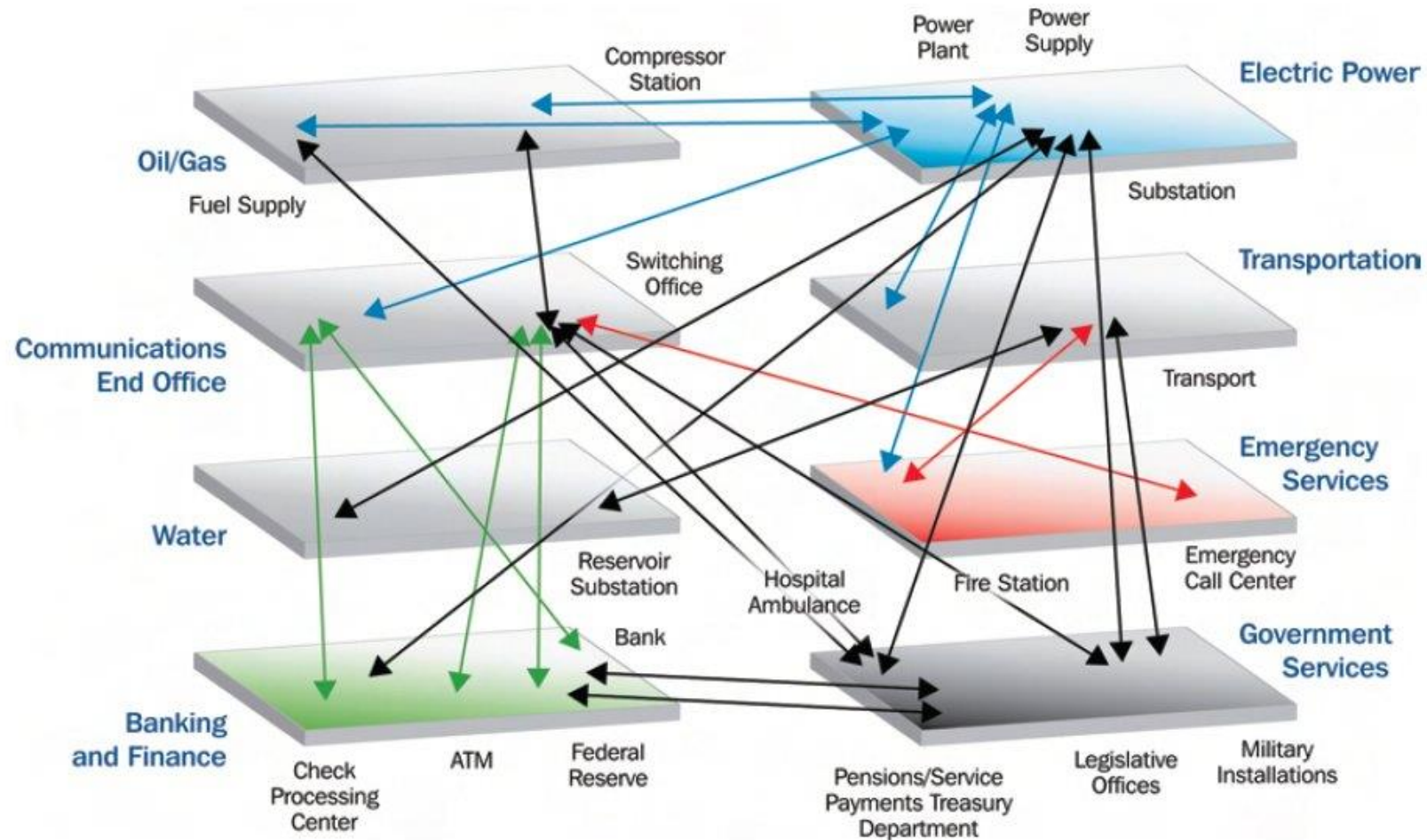


FIGURE 3.1 Connections and interdependencies across the economy. Schematic showing the interconnected infrastructures and their qualitative dependencies and interdependencies. SOURCE: Department of Homeland Security, National Infrastructure Protection Plan, available at http://www.dhs.gov/xprevprot/programs/editorial_0827.shtm.

Similar
Storms 1989
Quebec



Parker Solar Probe



Bhopal Disaster

2 December 1984

- 
- It felt like somebody had filled our bodies up with red chillies, our eyes had tears coming out, noses were watering, we had froth in our mouths. The coughing was so bad that people were writhing in pain. Some people just got up and ran in whatever they were wearing or even if they were wearing nothing at all. Somebody was running this way and somebody was running that way, some people were just running in their underclothes. People were only concerned as to how they would save their lives so they just ran.

- Champa Devi Shukla

A Survivor of Bhopal

WARNING: graphic photo on next slide

What Happened

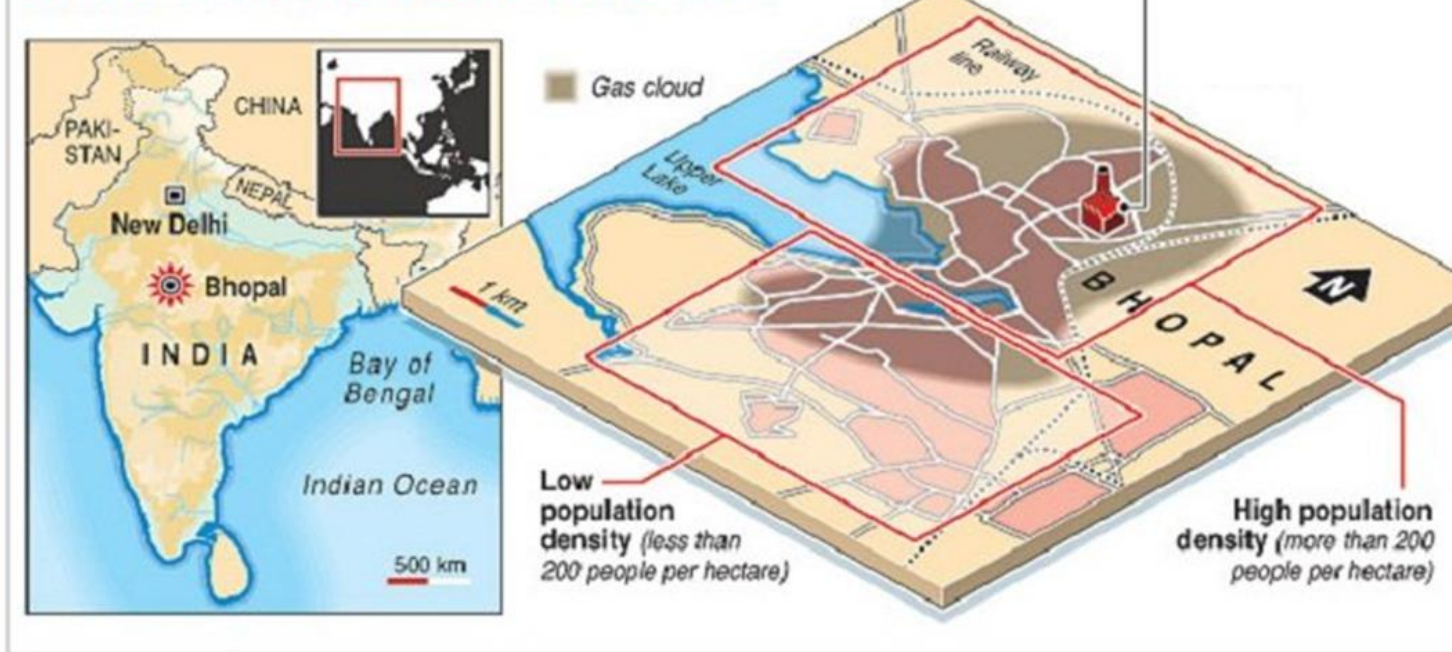
- Gas leak at the Union Carbide plant
- At the time of the leak, safety systems were not working, pipes and valves were in poor condition
- Indication that there was a leak at 11 pm
- Alarm system was triggered at 12:50 am
 - Two part alarm (plant and Public)
 - Public Alarm was turned off
 - Plant Employees evacuated upwind (out of the gas cloud)



Photo Credit: Raghu

BHOPAL

GAS LEAK DISASTER



- 42,000 kilograms of Methyl isocyanate and other gases
- Chemicals hugged the ground causing children and shorter people to be more effected

Gas Leak



Impacts on Society

- Immediate Death toll
 - 3783 official
 - 16,000+ claimed
- Over 500,000 non-fatal injuries
- Those exposed have 10x the cancer risk
- 43% of pregnancies did not survive
 - 14 of babies died within 30 days
 - Many that survived had birth defects

Long term impacts



- Ground water tests
 - Mercury levels 20, 000 and 6 millions times the expected levels
 - Chemicals known to cause cancer, brain damage, and birth defects were found at levels 50 times higher than the EPA safety limits
- Breast milk
 - Dangerous chemical transfer into the milk of mothers
- Continued Exposure to future generations

Implementing Resilience

What resilience strategies can be implemented to prevent this from happening again?





Space Shuttle Columbia

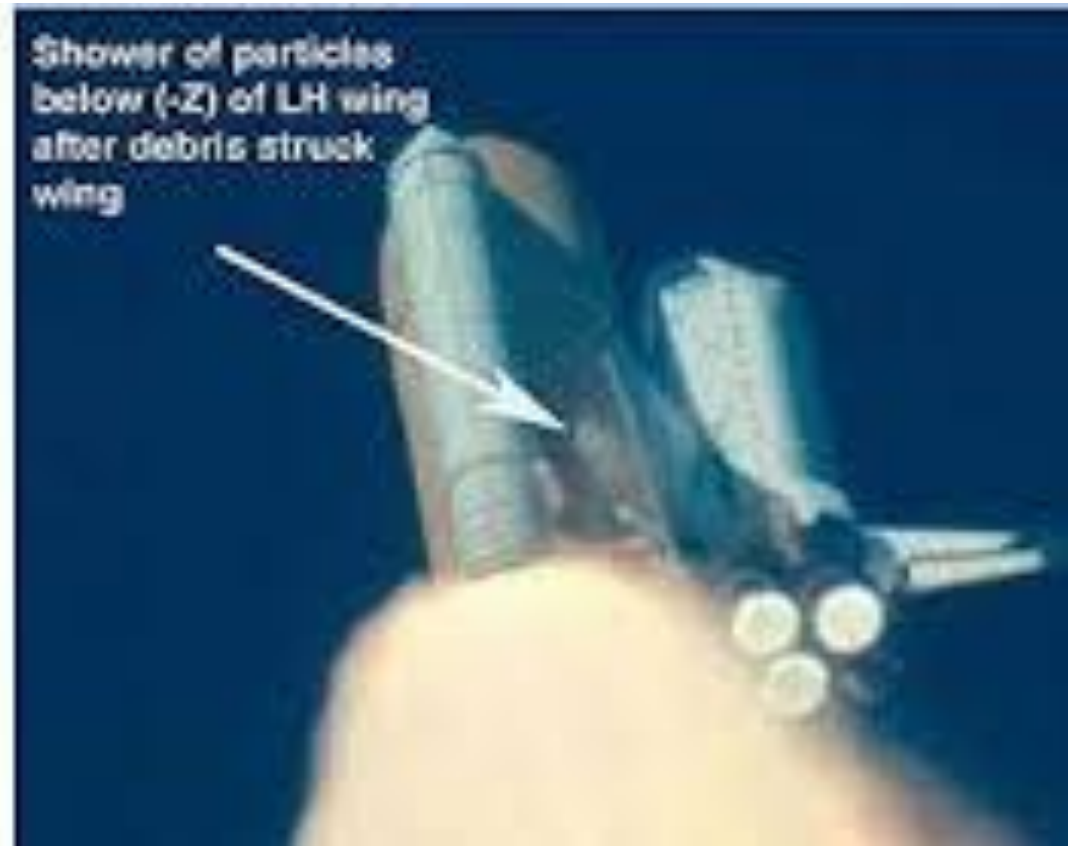
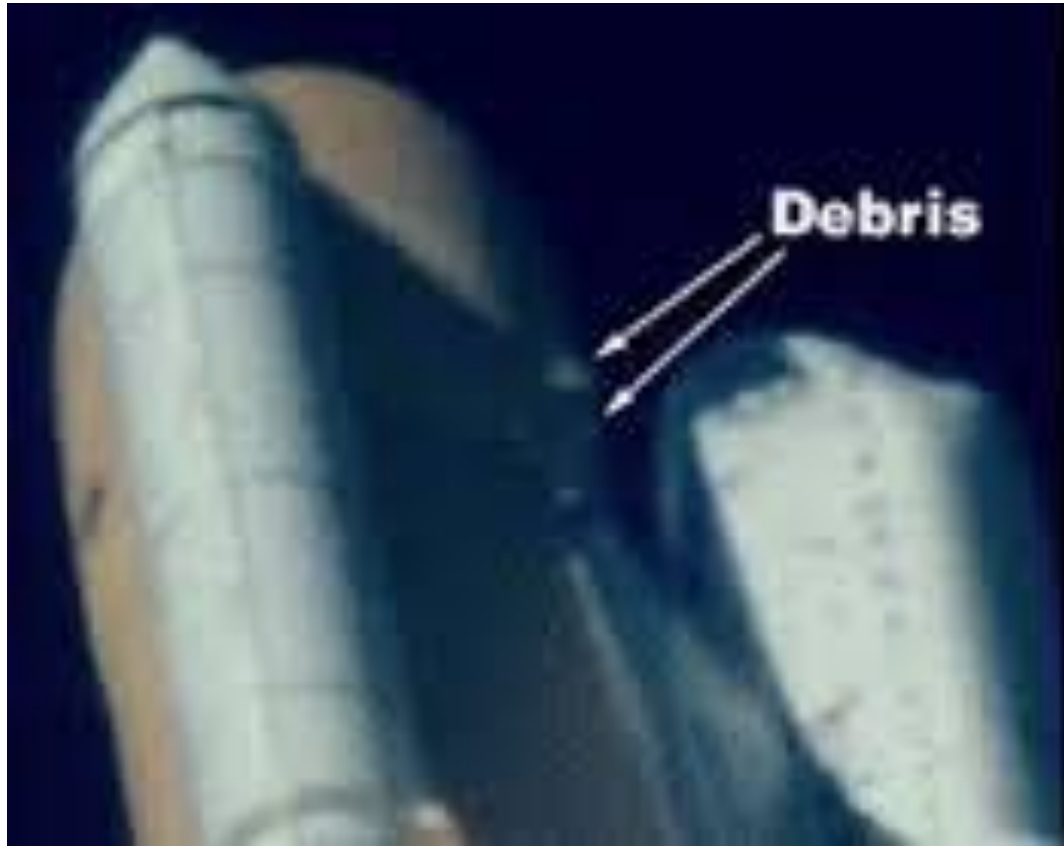
From left to right: David Brown, Rick Husband, Laurel Clark, Kalpana Chawla, Michael Anderson, William McCool, Ilan Ramon

Photo: NASA

What happened

- During launch, a piece of foam broke off of the external fuel tank and hit the left wing of the shuttle
- Engineers suspected that the damage was serious but limited the investigation because they didn't think there was anything the astronauts could do about it
- During re-entry, the damage allowed hot gas to enter and destroy the internal wing structure
- The orbiter became unstable and started to break apart

Cause of the accident



Example scenario – State Highway 94

- Hazards for the region?
- Vulnerabilities for the road?
- Methods to mitigate the hazards?
- In terms of the 4Rs of resilience?
 - Robustness
 - Redundancy
 - Resourcefulness
 - Rapidity



Machinery like this have been kept busy this winter clearing snow from Alpine sections of the Milford Road

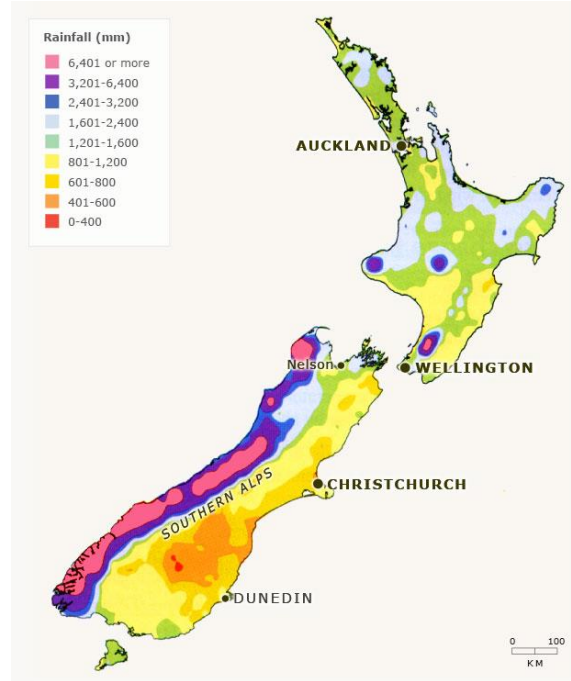
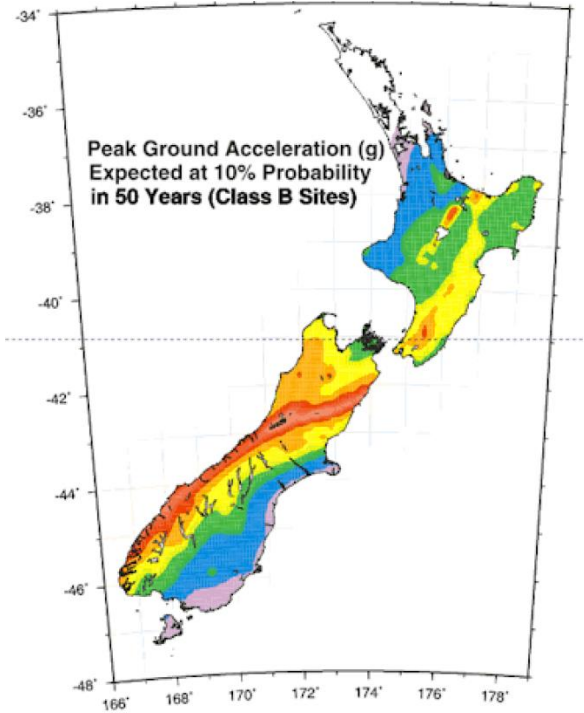


Heavy rain can scour loose rock and bring down slips like this on to Milford Road.



Weather station maintenance underway on a station located high above Milford Sound.

Example scenario – State Highway 94



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Example scenario – State Highway 94

Connections with other
infrastructure

Connections with the
community

Importance – local,
regional, and national

Community Resilience



- University of Waikato
 - What are the different components needed for a university?
- What can happen to the different components?
- What will happen if we lose a critical component?
- How can we prevent this?

Infrastructure Interconnectedness



- University of Waikato
 - What infrastructure systems does the university rely on for its day to day operations?
 - How does the university positively or negatively impact the infrastructure that it is tied into?
 - If the university were to expand, what considerations would need to be considered in terms of infrastructure?